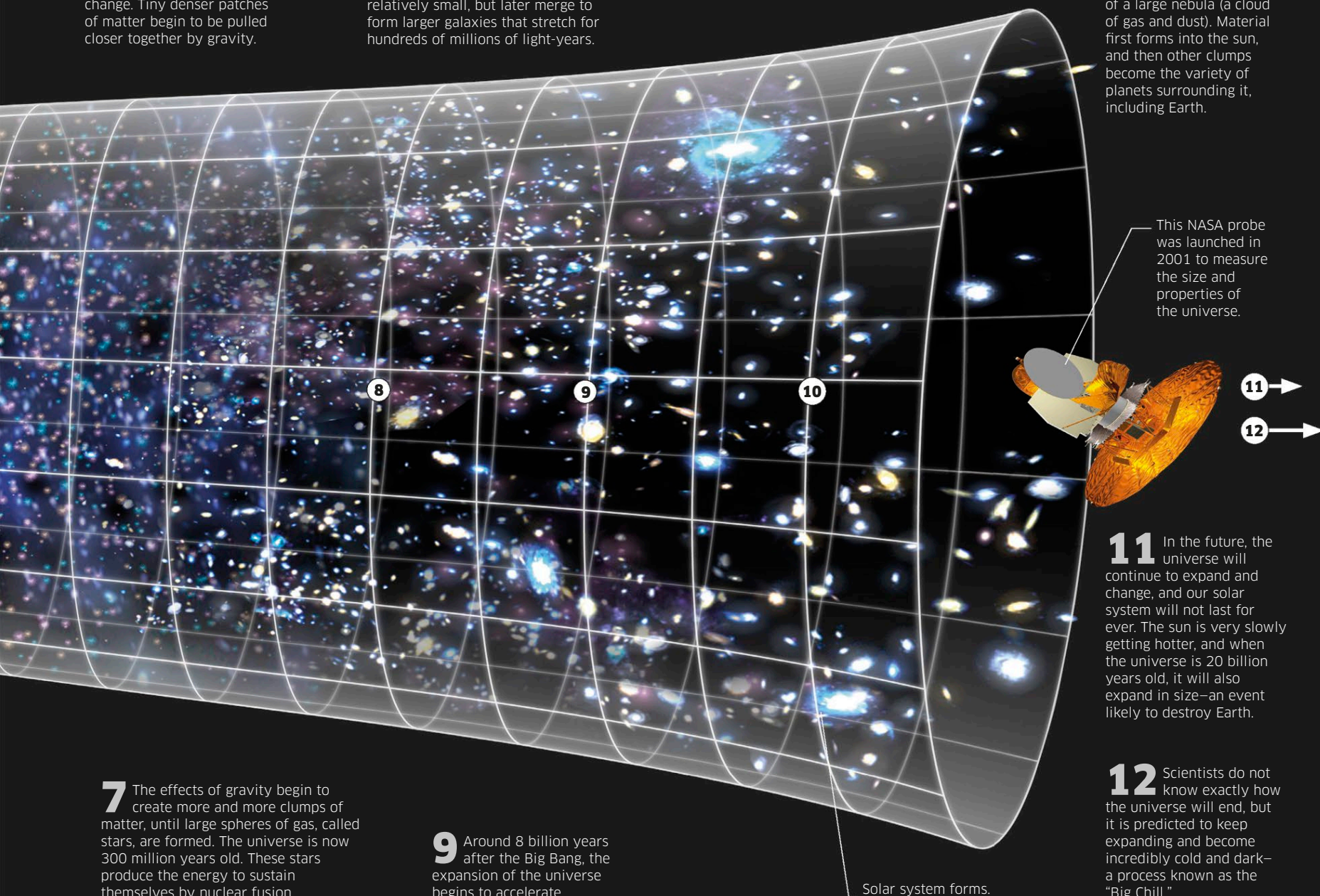


**6** Just over half a million years after the Big Bang, the distribution of matter in the universe begins to change. Tiny denser patches of matter begin to be pulled closer together by gravity.

**8** Stars form in groups within the universe's vast clouds of gas. The first groups become the first galaxies. Most of these are relatively small, but later merge to form larger galaxies that stretch for hundreds of millions of light-years.

**10** Our solar system comes into being after 9 billion years, formed from the collapse of a large nebula (a cloud of gas and dust). Material first forms into the sun, and then other clumps become the variety of planets surrounding it, including Earth.



**7** The effects of gravity begin to create more and more clumps of matter, until large spheres of gas, called stars, are formed. The universe is now 300 million years old. These stars produce the energy to sustain themselves by nuclear fusion.

**9** Around 8 billion years after the Big Bang, the expansion of the universe begins to accelerate.

Solar system forms.

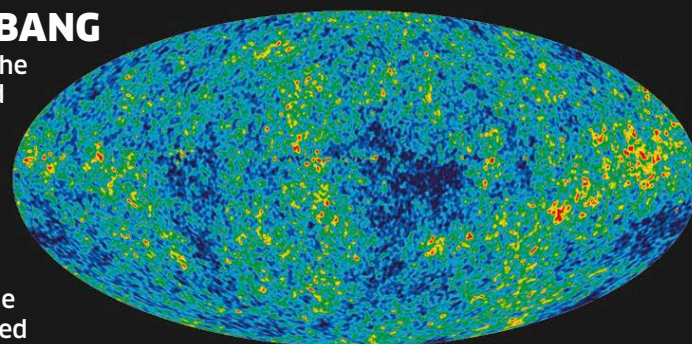
This NASA probe was launched in 2001 to measure the size and properties of the universe.

**11** In the future, the universe will continue to expand and change, and our solar system will not last for ever. The sun is very slowly getting hotter, and when the universe is 20 billion years old, it will also expand in size—an event likely to destroy Earth.

**12** Scientists do not know exactly how the universe will end, but it is predicted to keep expanding and become incredibly cold and dark—a process known as the “Big Chill.”

## DISCOVERING THE BIG BANG

Scientists did not always believe in the theory of an expanding universe and the Big Bang. However, during the 20th century, several discoveries were made which supported this idea. In 1929, American astronomer Edwin Hubble observed that the light coming from distant galaxies appeared redder than it should be. He attributed this to a phenomenon called redshift, suggesting that galaxies must be moving away from us. Another piece of evidence was the discovery of cosmic background radiation—microwaves coming from all directions in space that could only be explained as an after effect of the Big Bang.



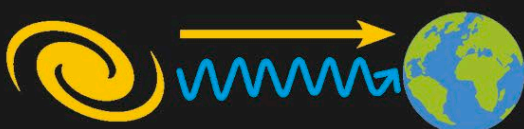
### Cosmic background radiation

This image, captured by NASA's Wilkinson Microwave Anisotropy Probe, shows a false color depiction of the background radiation that fills the entire universe. This is the remains of the intense burst of energy that was released by the Big Bang.



### Redshift

When an object (a distant galaxy) is moving away from the observer (us), its wavelengths get longer. The light it produces therefore shifts into the red end of the light spectrum. More distant galaxies have greater redshift—supporting the theory that the universe is expanding.



### Blueshift

A few nearby galaxies are actually moving toward us. Their wavelengths will be shorter, shifting the light they produce to the blue end of the spectrum.